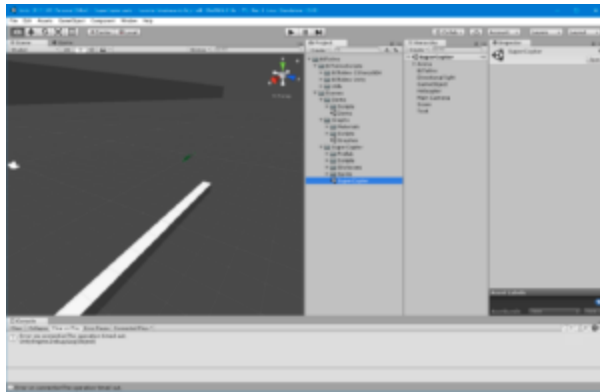


Tutoriel d'utilisation de l'API Bitalino avec Unity

Source : <https://www.creact.co.jp/category/item/measure/bitalino>



There are a few things to keep in mind when using the Unity API.

BLE module cannot be used in Unity at present.

If you really want to use it, you need to replace it with a Bluetooth module.

This is true for most other APIs. For details, [please refer to](#) the [previous explanation](#) .

Please consider BLE only for smartphones, especially Apple environments (iPhone, MacOSX).

Unity is also compatible with smartphones, but BITALino can only be used in the Windows environment so far.

This is because the method for communication varies depending on the OS, and the public API is created only for Windows.

Separately, if the communication part is made according to the environment, it may be somehow managed.

Finally, unfortunately, the currently released library is out of date.

Currently sold is BITALino (r) evolution, but there is a version without (r) evolution before, and it is made with that specification.

However, only the added PWM output cannot be used, and basic operations such as data acquisition will work.

As a preparatory step, Unity must be installed on the PC.

Unity is an integrated development environment, but when programming, a separate development environment (IDE) is required.

Previously, Unity came with an IDE called MonoDevelop, but Visual Studio can be used for a while.

Which one you use is up to you, but VisualStudio is recommended because it can be used for programming other than Unity.

Also, this is easier because you can optionally install it together when setting up VisualStudio.

There are many sites that explain how to do this. Please note that the installation time is very long.

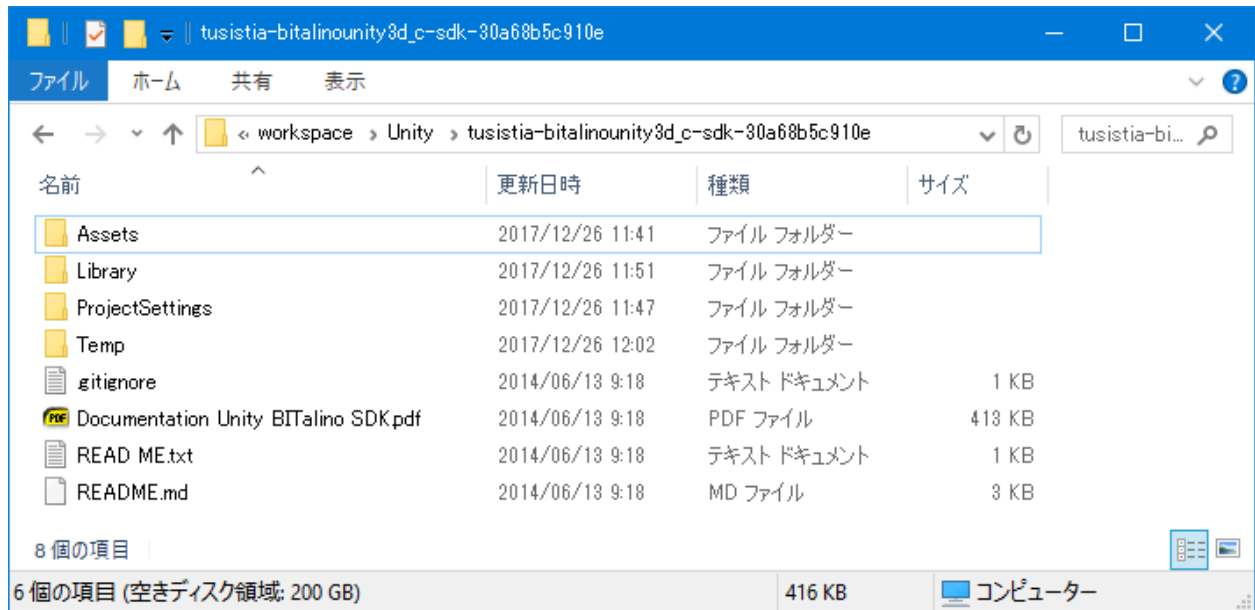
Have you already installed Unity? Use the API immediately.

First, download. You can download it from this page that was introduced before.

<http://bitalino.com/en/development/apis>

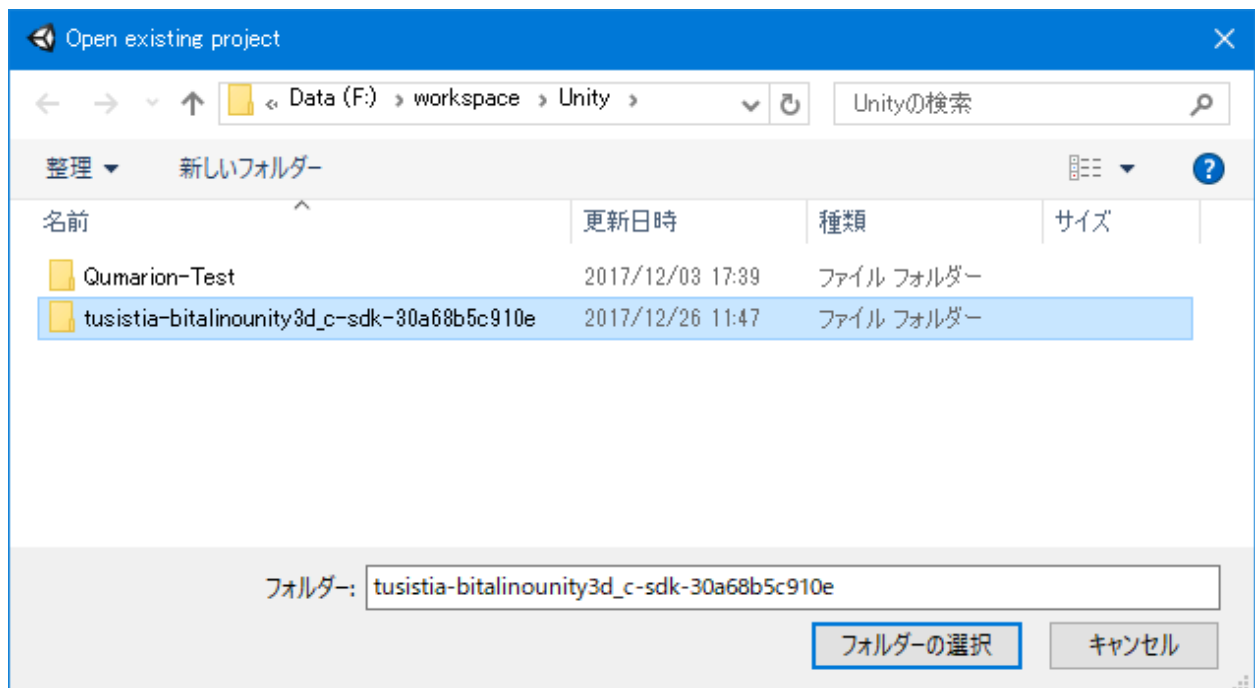
 RaspberryPi	C++ API					HTML	PL
	Python + PySerial Example						
 unity	Unity3D API Unity3D WF & WPF Example						Adrie LAF an
 Microsoft Visual C#	C# API						PL

Once downloaded, extract it to a suitable working folder.



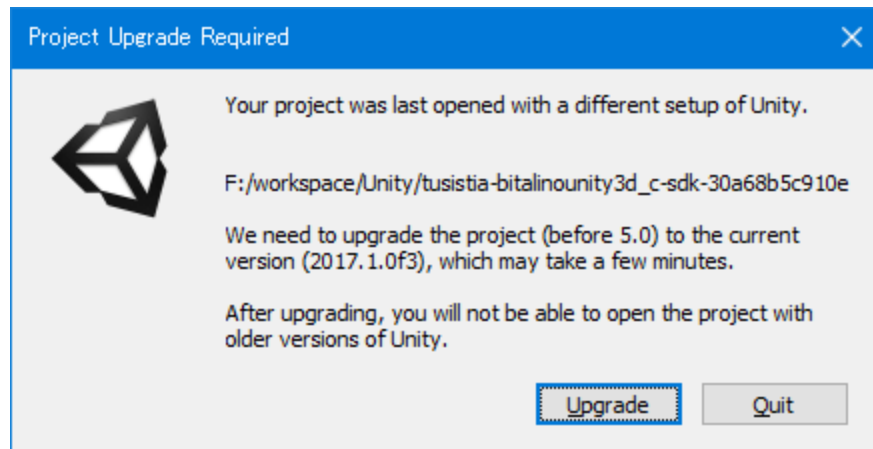
Then start Unity and open the project.

Unity sees the project as a “folder”, so with the folder you just expanded selected, click “Select Folder”.



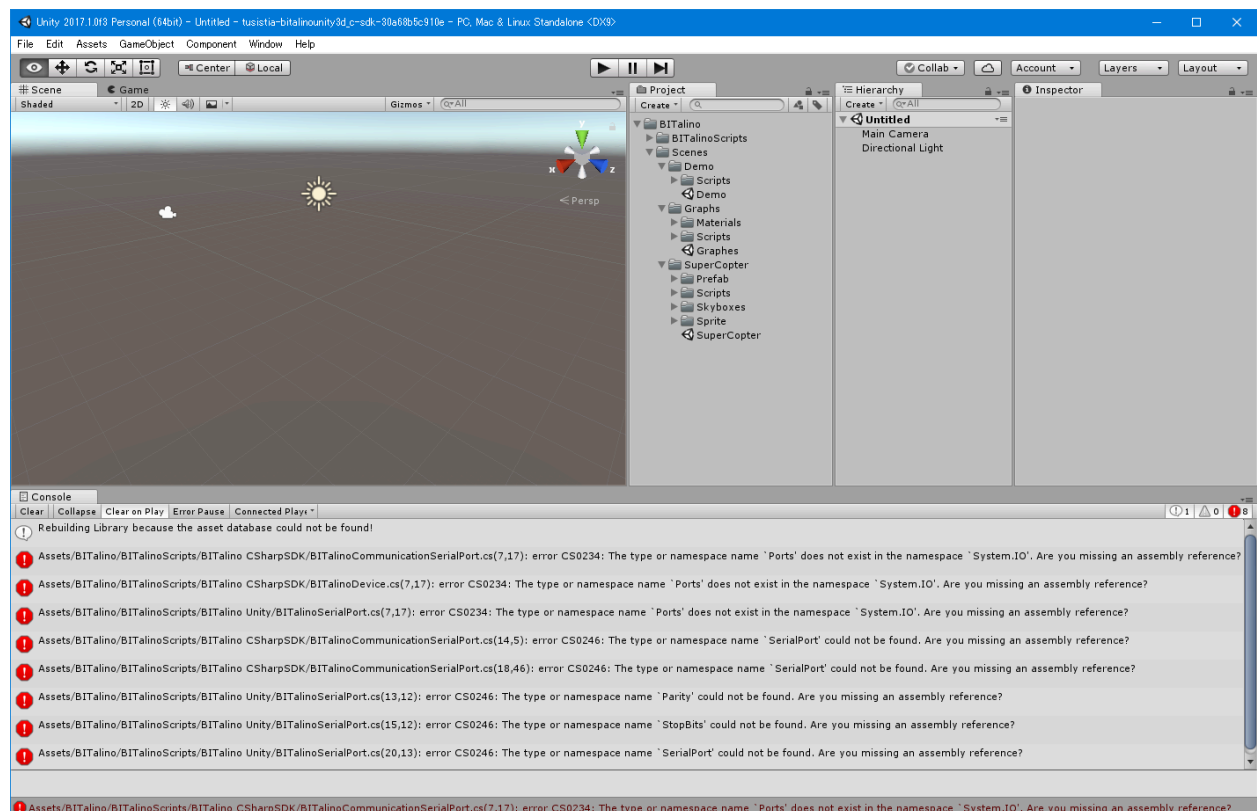
You may see a dialog like this while loading.

This is because the API is made with an older version of Unity, and I'm asking if I can upgrade. Press "Upgrade".



When the project has been loaded, the Unity screen will appear as shown in the image below.

I think that it works, but there are a lot of errors in the Console window.



Even if you press the play button as it is, you can't do anything because you are told "Correct all errors!"

By the way, it is a way to eliminate this error, but I am very sorry, but because of the time, I would like to comment on the next time.

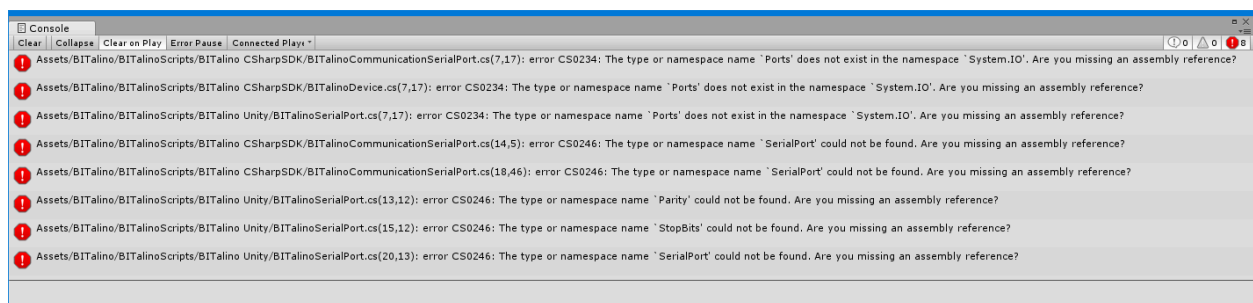
The correction itself does not require rewriting the code, it only changes one item.

If you check the error message, you will find a solution.

In order to make BITalino work with Unity, I explained the previous time to download the API and import it into Unity.

And I think that there were a lot of errors, but this time I will solve it.

First, let's look at the error message.



There are many things, but if you look closely, you will notice keywords related to something.

'Ports', 'SerialPort', 'Parity', 'StopBits' etc.

If you have touched an Arduino or other microcomputer, I think it will come with a pin.

BITalino communicates via Bluetooth, but the entity communicates via serial communication using a virtual COM port.

So, if you search for "Unity Serial Communication", there are many solutions.

The detailed contents are carefully explained by volunteers, so you can take a look there, but all you have to do is change the ".NET 2.0 Subset".

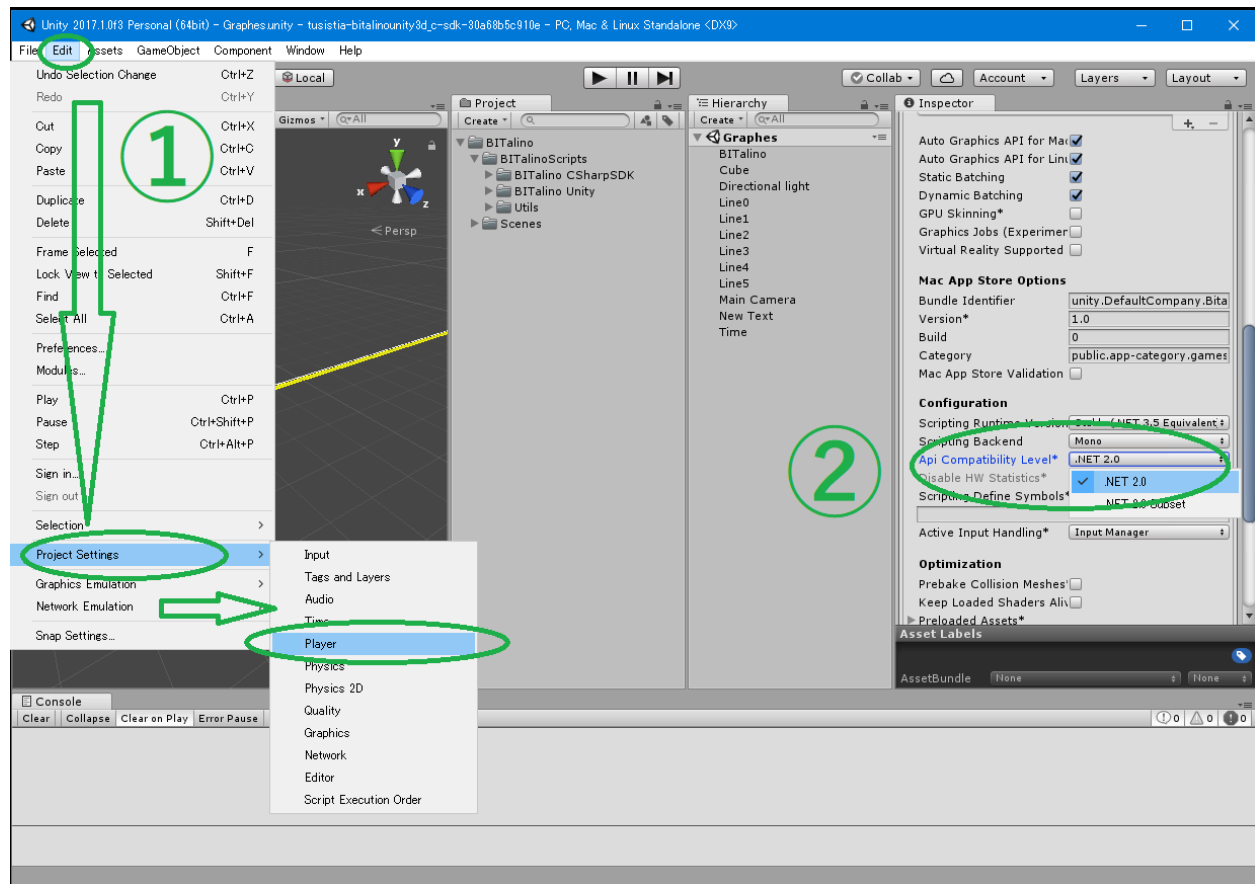
I'll briefly explain the procedure ...

First, select "Edit"-> "Project Settings"-> "Player" from the Unity menu bar.

The "Inspector" window will appear as PlayerSettings.

There are many items that can be set, but when you scroll to the middle, there is an item called "Api Compatibility Level", and ".NET 2.0 Subset" is set.

You can change this to ".NET 2.0".



What is that? The error disappears beautifully.

Now, I think that I can finally run BITALino, but I still have my own settings.

Continuing from the previous time, this time we will configure BITALino. Go to where you want to move.

I have put together all the images in the lower part, but I will supplement them.

(1) Open the "Graphes" scene from the Project window.

The API includes three sample scenes.

In the second scene "Graphes", you can draw a graph on the screen as the name suggests. The third "SuperCopter" seems to be game-like, but I couldn't understand how to play it. . .

② The game objects in the scene are displayed in the Hierarchy window. Select the “BITalino” object from them.

The Inspector window displays the script attached to the BITALino object.

There are a lot of items, but there are 3 places to set.

(3) “PortName” is the name of the COM port to be used. Check from PC Device Manager.

When BITalino pairing is complete, two serial ports will be added to “Ports (COM and LPT)”.

The reason why there are two is like the Bluetooth specification. For the time being, set the number of the younger one and try the other one if it doesn't work.

④ “Sampling Rate” sets how many times data is acquired per second. The value that can be set here is 1,10,100,1000. Do not set any other value. The default value is 1000, but since it may become unstable, leave it at 100 for the time being.

(5) There is an item “Buffer Size” at the bottom. Set this to about twice the “Sampling Rate” you set earlier.

If you set a value that is too far away, it will take time until the buffer is filled, and Unity will freeze.

⑥ There are 3 scripts to use :

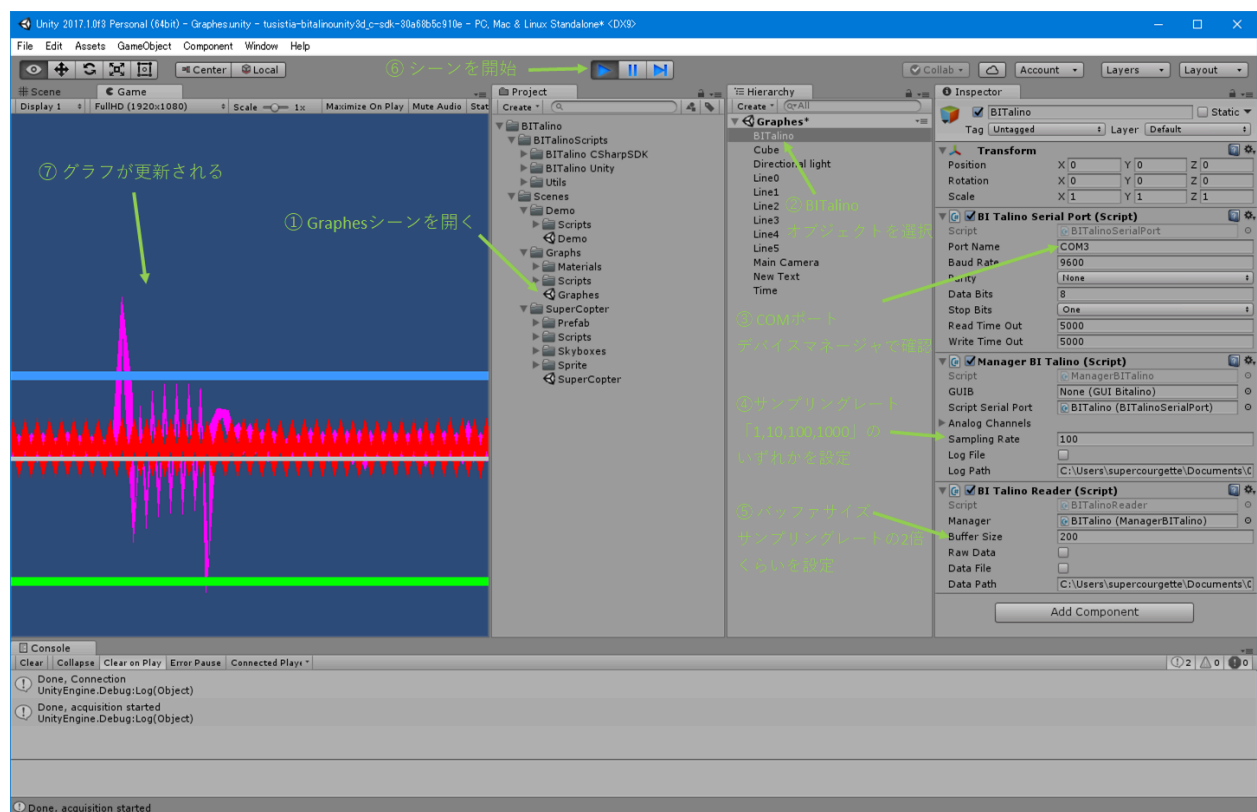
- BitalinoManager
- BitalinoReader
- BitalinoSerialPort

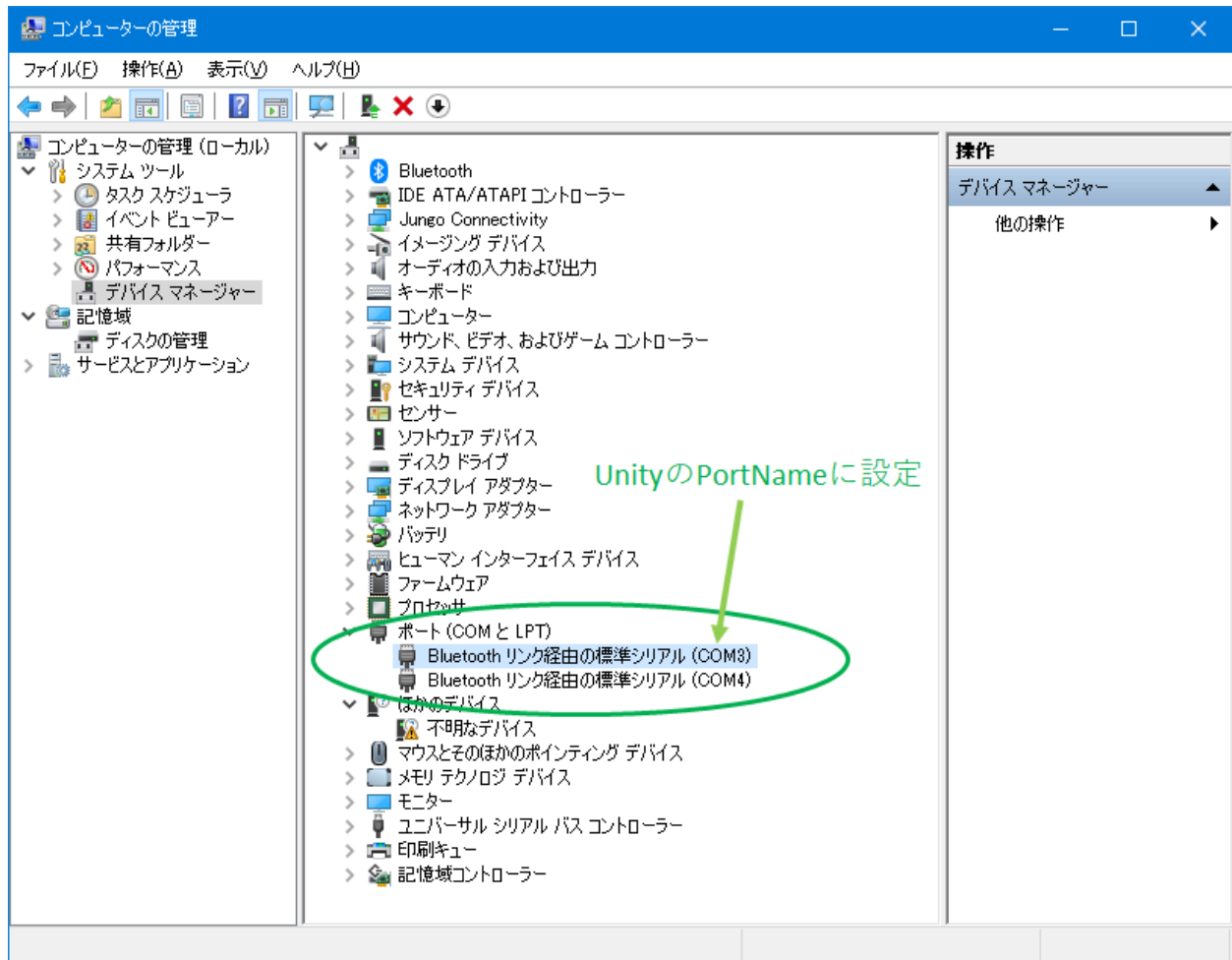
To launch, you can use the Graphs scene, with data checked with RawData or not.

⑦ After completing the settings, press the play button to start the scene.

(8) The LED of BITALino starts to flash quickly, and after a while, the graph moves and updates.

If the buffer size is large, it will take some time to update.





Did you move well?

If there is a problem, an error message will be displayed in the Console window. Check the contents and investigate the cause.

I have confirmed the operation for the time being, so I would like to use other APIs next time.

The details on how to use the UnityAPI may be explained somewhere, but first try it out in various ways.

Tips [thanks to C. Bourgue - ANDROIDE 2021]

- Check for locks that may be due to the antivirus
- Check that, if there are several bluetooth devices, leave only the bluetooth of the Bitalino unit
- Check that the sensor wires are all fully connected
- Make sure you don't have false contacts